

Separating Mixtures — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. On this topic the method is the cheap mark and the property that differs is the expensive one: insist on "because the sand is insoluble", "because water has the lower boiling point", "because iron is magnetic". Every R_f on these papers was computed in the generator, not typed; any value above 1 in a candidate's work is the division done upside down. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 **filtration; evaporation** (or crystallisation); **simple distillation; magnet; chromatography** [5]
One mark each.
Rows 2 and 3 are the same mixture with a different aim. A candidate who writes the same method for both has not asked "which part do I want?"
- 2 (a) **residue** [1] [2]
(b) **filtrate** [1]
- 3 (a) mud (residue) on the paper; water (filtrate) passes through [1] [3]
(b) the mud particles are larger than the holes in the filter paper, so they cannot pass [1]; water particles are much smaller than the holes, so they go through [1]
- 4 Funnel with cone of filter paper, standing in a conical flask [1] [4]
Filter paper and funnel labelled [1]
Residue labelled on the paper [1]
Filtrate labelled in the flask [1]
- 5 Pour the solution into an evaporating dish and heat it (gently, over a beaker of water or a tripod) [1] [3]
The water evaporates / turns to vapour and leaves [1]
The salt cannot evaporate and is left behind as crystals; dry them [1]
- 6 (a) Use a **magnet** [1] [3]
(b) Iron is magnetic and is attracted to the magnet [1]; sand is not magnetic and stays behind [1]
- 7 The salt has dissolved [1] [3]
so its particles are small enough to pass through the holes in the filter paper [1]
along with the water — filtration only removes insoluble solids [1]
- 8 Evaporation lets the water escape into the air as vapour, so it is lost — only the salt is kept [1] [3]
To collect the water it must be condensed back into a liquid [1]
so **simple distillation** is the right method [1]
- 9 Baseline in pencil [1] — ink is soluble and would run up the paper with the sample [1] [4]
Solvent level below the baseline [1] — otherwise the spots dissolve into the solvent instead of travelling up the paper [1]
Also accept: small concentrated spots (so they do not smear); lid on the container (stops solvent evaporating); mark the solvent front immediately (it fades).

Paper B — Medium

30 MARKS

- 1 (a) The water boils (at 100 °C) and turns into vapour / steam [1] [7]
 (b) To turn the vapour back into liquid (condense it) [1]; cold water flows around the outside, cooling the vapour below its boiling point [1]
 (c) **Distillate** (accept: pure water) [1]
 (d) The salt [1]; its boiling point is far higher, so it did not turn into vapour at 100 °C [1]
 (e) **100 °C** [1]
- 2 So the condenser is always full of cold water [1] [2]
 and the coldest water is nearest the outlet, so the vapour meets colder and colder walls as it travels and is fully condensed by the end [1]
- 3 Ink is a mixture of different dyes [1] [3]
 The dyes differ in how soluble they are in the solvent (and how strongly they are attracted to the paper) [1]
 so as the solvent moves up it carries the more soluble dyes further, and they separate into spots at different heights [1]
- 4 (a) E102: $2.0 \div 8.0 = 0.25$ [1]; E133: $6.8 \div 8.0 = 0.85$ [1] [5]
 (b) **E102 and E133** [1]; its spots are at the same heights (same R_f) as those dyes, and the same substance always travels the same distance in the same solvent [1]
 (c) It gives two spots, so it contains at least two substances [1]
Working must be shown in (a). $8.0 \div 2.0 = 4$ is the upside-down version — no marks, and it is impossible because R_f cannot exceed 1.
- 5 $9.0 \div 12.0$ [1] [2]
 = **0.75** (no units) [1]
- 6 Add water and stir [1] — the salt dissolves and the sand does not [1] [6]
 Filter [1] — the sand is the residue; the salt solution is the filtrate [1]
 Evaporate the filtrate in an evaporating dish [1] — the water leaves and dry salt crystals remain [1]
The reasons are half the marks. A bare list of three steps scores three.
- 7 The salt must be dissolved before filtering so that it can pass through and be separated [3]
 from the sand [1]
 If evaporated first, the water would leave and the salt would be deposited back onto the sand [1]
 leaving salty sand — nothing separated [1]
- 8 Separating funnel (accept decanting) [1] [2]
 Oil and water do not mix, and the oil floats on top (less dense) [1]

Paper C — Hard

30 MARKS

- 1 (a) Evaporation / crystallisation [1]; the copper sulfate is dissolved so cannot be filtered [1]; water has a much lower boiling point and leaves, the solid stays [1] [9]
 (b) Filtration [1]; chalk is insoluble, so its particles are large enough to be trapped by the paper [1]; water passes through [1]
 (c) (Simple) distillation [1]; the two liquids have different boiling points [1]; ethanol boils first at 78 °C, is condensed and collected, water is left behind [1]
(c) is beyond what the page teaches directly and is meant to be: the boiling-point property is the same one as for sea water. A candidate who answers "fractional distillation" with the same reasoning gets full marks.

- 2 (a) Blue: $4.5 \div 7.5 = \mathbf{0.6}$ [1]; yellow: $1.5 \div 7.5 = \mathbf{0.2}$ [1] [6]
 (b) **Yes** [1]; the yellow spot is at the same height / same R_f as the Tartrazine spot [1]
 (c) The blue dye is more soluble in the solvent (or less attracted to the paper) [1]; so the solvent carries it further before it is left behind [1]
- 3 The spot is carried by the solvent, so it can never travel further than the solvent front [1] [3]
 so spot distance $\div$ solvent distance is always less than (or at most equal to) 1 [1]
 1.5 means the student divided the solvent distance by the spot distance — the wrong way round [1]
- 4 Ink baseline: the ink is soluble, so the line dissolves and moves up the paper [1], [4]
 mixing with the sample and making the spots impossible to read [1]
 Solvent above baseline: the spots are under the solvent and dissolve into it [1], so they are washed off instead of travelling up the paper [1]
- 5 It reads above 100 °C [1] [3]
 because it measures the boiling point of the whole solution, which is raised by the dissolved dyes [1]
 so the student cannot tell from the thermometer that pure water is what is coming over [1]
- 6 The salt is dissolved, so its particles pass through the filter paper with the water [1] [3]
 Filtration only removes insoluble solids [1]
 She should evaporate the water (or crystallise) to obtain the salt [1]
- 7 Distillation has to boil every bit of the liquid — turning it all into vapour takes a great [2]
 deal of energy [1]
 Filtration just lets the liquid run through the paper; nothing has to change state [1]

Paper D — Hard

30 MARKS

- 1 Magnet first [1] — iron is magnetic, the others are not; and it must be done while [7]
 everything is dry [1]; now: iron on the magnet, sand and salt left
 Add water and stir [1] — salt dissolves, sand does not [1]
 Filter [1] — sand is the residue (rinse and dry it), salt solution is the filtrate
 Evaporate the filtrate [1] — water leaves, dry salt crystals remain [1]
The order mark is for the magnet coming before the water. After the water, the iron would be wet, rusting, and mixed into the residue with the sand.
- 2 (a) distance = $R_f \times$ solvent distance = 0.80×9.0 [1] = **7.2 cm** [1] [5]
 (b) $3.6 \div 9.0$ [1] = **0.4** [1]
 (c) R_f is fixed for a given dye in a given solvent, so it identifies the dye as long as it is compared with a value measured in the same solvent [1]
(a) is R_f rearranged, which most candidates have never done. Award the first mark for the rearrangement even if the arithmetic slips.

- 3** Award one mark for each of these, up to six: [6]
- The sea water is heated and the water boils at 100 °C, turning to vapour.
 The salt has a much higher boiling point and stays in the flask (the solution becomes more concentrated).
 The vapour passes into the condenser, which has cold water flowing round the outside.
 The vapour is cooled below its boiling point and condenses back to liquid water.
 The thermometer at the top of the flask reads 100 °C while water is coming over, showing what is being collected.
 The collected distillate is pure because only the water turned to vapour; nothing dissolved could travel with it.
- For full marks the answer must say why the product is pure (only water vaporised), not just that it "is pure". Award at most four if the condenser is described without saying what it does to the vapour.*
- 4 Evaporation [1]** [3]
- The water has turned to vapour and escaped into the air [1]
 The salt cannot evaporate at that temperature (far higher boiling point), so it is left behind as crystals [1]
- 5** Chromatography does not collect anything — the water soaks into the paper and evaporates [1] [3]
 To obtain the water you would need distillation [1]
 Chromatography would show how many dyes the ink contains and, with R_f values, which ones [1]
- 6** (i) The mistake: dissolved salt passes straight through filter paper — filtration only catches insoluble solids [1] [6]
 Correct version: evaporate the water from the solution (or crystallise); the salt is left behind in the dish [2]
 (ii) The mistake: the division is upside down, which would give values above 1 [1]
 Correct version: $R_f = \text{distance moved by the spot} \div \text{distance moved by the solvent}$, both measured from the baseline [2]
- These are the two most common wrong sentences in the whole topic. If he can mark them in someone else's work, he is unlikely to write them in his own.*